1) a)

$\lim_{x\to\infty}(1+\mathrm{e}^{-x})=1$

1) b)

$\lim_{x\to 0}(x^4\mathrm{e}^{x})=0$

2) a)

Since $4\times 0^3+1\neq 0$ then

$\lim_{x\to 0}\left(\sqrt{\frac{x^3-2x^2+10}{4x^3+1}}\right)=\lim_{x\to 0}\left(\sqrt{\frac{0-0+10}{0+1}}\right)=\sqrt{10}$

2) b)

Since $3\times 0^2-0+2\neq 0$ then

$\lim_{x\to 0}\left(\frac{x+4}{\sqrt{3x^2-5x+2}}\right)=\lim_{x\to 0}\left(\frac{0+4}{\sqrt{0-0+2}}\right)=\frac{4}{\sqrt{2}}=2\sqrt{2}$

2) c)

Since $(0)^2-2\neq 0$ then

$\lim_{x\to 0}\left(\sqrt{\frac{2x^3-5x^2-1}{x^2-2}}\right)=\lim_{x\to 0}\left(\sqrt{\frac{0-0-1}{0-2}}\right)=\frac{1}{\sqrt{2}}=\frac{\sqrt{2}}{2}$

3) a)

$\lim_{n\to\infty}\left(\frac{5-4n}{n}\right)=\lim_{n\to\infty}\left(\frac{\frac{5}{n}-\frac{4n}{n}}{\frac{n}{n}}\right)=\lim_{n\to\infty}\left(\frac{5}{n}-4\right)=-4$

3) b)

$\lim_{n\to\infty}\left(\frac{2n-3}{n^2}\right)=\lim_{n\to\infty}\left(\frac{\frac{2n}{n^2}-\frac{3}{n^2}}{\frac{n^2}{n^2}}\right)=\lim_{n\to\infty}\left(\frac{2}{n}-\frac{3}{n^2}\right)=0$

3) c)

$\lim_{n\to\infty}\left(\frac{n+7}{2n+1}\right)=\lim_{n\to\infty}\left(\frac{1+\frac{7}{n}}{2+\frac{1}{n}}\right)=\frac{1}{2}$

3) d)

$$\lim_{n\to\infty}\left(\frac{10-3n}{n-3}\right)=\lim_{n\to\infty}\left(\frac{\frac{10}{n}-3}{1-\frac{3}{n}}\right)=-3$$

4) a)

$$\lim_{n\to\infty}\left(\frac{2n^2-4n+5}{3n^2+n-4}\right)=\lim_{n\to\infty}\left(\frac{\frac{2n^2}{n^2}-\frac{4n}{n^2}+\frac{5}{n^2}}{\frac{3n^2}{n^2}+\frac{n}{n^2}-\frac{4}{n^2}}\right)=\lim_{n\to\infty}\left(\frac{2-\frac{4}{n}+\frac{5}{n^2}}{3+\frac{1}{n}-\frac{4}{n^2}}\right)=\frac{2}{3}$$

4) b)

$$\lim_{n\to\infty}\left(\frac{5n^2+2n+1}{2n^2-n-7}\right)=\lim_{n\to\infty}\left(\frac{\frac{5n^2}{n^2}-\frac{2n}{n^2}+\frac{1}{n^2}}{\frac{2n^2}{n^2}-\frac{n}{n^2}-\frac{7}{n^2}}\right)=\lim_{n\to\infty}\left(\frac{5-\frac{2}{n}+\frac{1}{n^2}}{2-\frac{1}{n}-\frac{7}{n^2}}\right)=\frac{5}{2}$$

4) c)

$$\lim_{n\to\infty}\left(\frac{5n^2-4n+5}{4-n-7n^2}\right)=\lim_{n\to\infty}\left(\frac{\frac{5n^2}{n^2}-\frac{4n}{n^2}+\frac{5}{n^2}}{\frac{4}{n^2}-\frac{n}{n^2}-\frac{7n^2}{n^2}}\right)=\lim_{n\to\infty}\left(\frac{5-\frac{4}{n}+\frac{5}{n^2}}{\frac{4}{n^2}-\frac{1}{n}-7}\right)=-\frac{5}{7}$$

5) a)

$$\lim_{x\to2}\left(\frac{x^2-4}{x-2}\right)=\lim_{x\to2}\left(\frac{(x-2)(x+2)}{x-2}\right)$$

But since $x\neq2$ then

$$\lim_{x\to2}\left(\frac{(x-2)(x+2)}{x-2}\right)=\lim_{x\to2}(x+2)=4$$

5) b)

$$\lim_{x\to1}\left(\frac{x^3-1}{x-1}\right)=\lim_{x\to1}\left(\frac{(x-1)(x^2+x+1)}{x-1}\right)$$

But since $x\neq1$ then

$$\lim_{x\to1}(x^2+x+1)=1+1+1=3$$

6) a)

$$\lim_{x\to3}\left(\frac{\sqrt{x}-\sqrt{3}}{x-3}\right)=\lim_{x\to3}\left(\frac{\sqrt{x}-\sqrt{3}}{x-3}\times\frac{\sqrt{x}+\sqrt{3}}{\sqrt{x}-\sqrt{3}}\right)=\lim_{x\to3}\left(\frac{x-3}{(x-3)(\sqrt{x}+\sqrt{3})}\right)$$

But since $x\neq3$ then

$$\lim_{x\to3}\left(\frac{x-3}{(x-3)(\sqrt{x}+\sqrt{3})}\right)=\lim_{x\to3}\left(\frac{1}{\sqrt{x}+\sqrt{3}}\right)=\frac{1}{2\sqrt{3}}$$

6) b)

$$\lim_{x\to 4}\left(\frac{\sqrt{x}-2}{x-4}\right)=\lim_{x\to 4}\left(\frac{\sqrt{x}-2}{x-4}\times\frac{\sqrt{x}+2}{\sqrt{x}+2}\right)=\lim_{x\to 4}\left(\frac{x-4}{(x-4)(\sqrt{x}+2)}\right)$$

But since $x\neq 4$ then

$$\lim_{x\to 4}\left(\frac{x-4}{(x-4)(\sqrt{x}+2)}\right)=\lim_{x\to 4}\left(\frac{1}{\sqrt{x}+2}\right)=\frac{1}{4}$$

6) c)

$$\lim_{x\to k}\left(\frac{\sqrt{k}-\sqrt{x}}{x-k}\right)=\lim_{x\to k}\left(\frac{\sqrt{k}-\sqrt{x}}{x-k}\times\frac{\sqrt{k}+\sqrt{x}}{\sqrt{k}-\sqrt{x}}\right)=\lim_{x\to k}\left(\frac{k-x}{(x-k)(\sqrt{k}+\sqrt{x})}\right)$$

But since $x\neq k$ then

$$\lim_{x\to k}\left(\frac{k-x}{(x-k)(\sqrt{k}+\sqrt{x})}\right)=\lim_{x\to k}\left(\frac{-1}{\sqrt{k}+\sqrt{x}}\right)=-\frac{1}{2\sqrt{k}}$$

7) a)

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)\approx\lim_{x\to 0}\left(\frac{x}{x}\right)=\lim_{x\to 0}1=1$$

7) b)

$$\lim_{x\to 0}\left(\frac{\cos x-1}{x}\right)\approx\lim_{x\to 0}\left(\frac{1-\frac{x^2}{2}-1}{x}\right)=\lim_{x\to 0}\left(\frac{-\frac{x^2}{2}}{x}\right)=\lim_{x\to 0}\left(-\frac{x}{2}\right)=0$$

8) a)

$$\lim_{x\to 0}\left(\frac{\sin 3x}{x}\right)\approx\lim_{x\to 0}\left(\frac{3x}{x}\right)=\lim_{x\to 0}(3)=3$$

8) b)

$$\lim_{x\to 0}\left(\frac{1-\cos 4x}{x^2}\right)\approx\lim_{x\to 0}\left(\frac{1-\left(1-\frac{(4x)^2}{2}\right)}{x^2}\right)=\lim_{x\to 0}\left(\frac{8x^2}{x^2}\right)=\lim_{x\to 0}(8)=8$$

8) c)

$$\lim_{x\to 0}\left(\frac{4\cos x}{\sin x+\cos x}\right)\approx\lim_{x\to 0}\left(\frac{4\left(1-\frac{x^2}{2}\right)}{x+1-\frac{x^2}{2}}\right)=\lim_{x\to 0}\left(\frac{4-2x^2}{x+1-\frac{x^2}{2}}\right)=4$$

9)

$$\lim_{x\to 0}\left(\frac{8\ln\left(1+\frac{x}{2}\right)}{x}\right)\approx\lim_{x\to 0}\left(\frac{8\left(\frac{x}{2}-\frac{x^2}{8}+\frac{x^3}{24}\right)}{x}\right)=\lim_{x\to 0}\left(\frac{4x-x^2+\frac{x^3}{3}}{x}\right)=\lim_{x\to 0}\left(4-x+\frac{x^2}{3}\right)=4$$

10)

$$\lim_{x\to 0}\left(\frac{3x\cos 2x-\sin 3x}{3x^3}\right)\approx\lim_{x\to 0}\left(\frac{3x(1-2x^2)-\left(3x-\frac{9}{2}x^3\right)}{3x^3}\right)=\lim_{x\to 0}\left(\frac{-\frac{3}{2}x^3}{3x^3}\right)=\lim_{x\to 0}\left(-\frac{1}{2}\right)=-\frac{1}{2}$$

11)

$$(4+x)^{\frac{1}{2}}=2\left(1+\frac{x}{4}\right)^{\frac{1}{2}}=2\left(1+\frac{1}{8}x-\frac{1}{128}x^2+\frac{1}{1024}x^3\right)=2+\frac{1}{4}x-\frac{1}{64}x^2+\frac{1}{512}x^3$$

So

$$\lim_{x\to 0}\left(\frac{-2+\sqrt{x+4}}{x}\right)\approx\lim_{x\to 0}\left(\frac{-2+2+\frac{1}{4}x-\frac{1}{64}x^2+\frac{1}{512}x^3}{x}\right)=\lim_{x\to 0}\left(\frac{\frac{x}{4x}-\frac{x^2}{64x}+\frac{x^3}{512x}}{\frac{x}{x}}\right)=\lim_{x\to 0}\left(\frac{1}{4}-\frac{x}{64}+\frac{x^2}{512}\right)=\frac{1}{4}$$